

UNIVERSITY ALUMNI MANAGEMENT SYSTEM

Development, Design and System Architecture Report

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Developer: Student Project Team

Technologies Used

- Vite.js
- Tailwind CSS
- JavaScript
- HTML5
- CSS3

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1. INTRODUCTION

The University Alumni Management System is a web-based platform developed to improve communication, networking, and engagement between alumni, students, and the university administration. The system provides a centralized digital environment where graduates can create professional profiles, share job opportunities, participate in university events, and mentor current students.

Traditional alumni management approaches often rely on manual records and fragmented communication channels, which limit collaboration and networking opportunities. This project addresses these challenges by introducing an interactive and scalable web solution developed using modern frontend technologies such as Vite.js and Tailwind CSS.

The system is designed to enhance alumni participation, simplify information management, and strengthen relationships between graduates and the university community.

2. PROBLEM STATEMENT

Many universities face difficulties in maintaining communication with alumni after graduation. Existing systems are often outdated, decentralized, or entirely manual, making it difficult to track alumni activities, provide mentorship opportunities, and organize networking events.

Students also lack accessible platforms for connecting with experienced alumni who can guide them academically and professionally.

Therefore, there is a need for a modern web-based system that can:

- Maintain updated alumni records
 - Facilitate networking and mentorship
 - Share employment opportunities
 - Improve university-alumni engagement
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3. OBJECTIVES

3.1 Main Objective

To develop a web-based University Alumni Management System that improves communication, networking, and engagement between alumni, students, and the university.

3.2 Specific Objectives

- To enable alumni registration and profile management
 - To allow sharing of job opportunities and events
 - To provide mentorship connections between alumni and students
 - To create a centralized alumni database
 - To improve communication between the university and graduates
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4. SCOPE OF THE SYSTEM

The system focuses on:

- Alumni registration and login
- Alumni profile management
- Event announcements
- Job posting and applications
- Student-alumni mentorship
- Admin management dashboard

The system does not include:

- Financial transactions
 - Academic grading systems
 - Student admission processing
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5. SYSTEM REQUIREMENTS

5.1 Functional Requirements

The system shall:

- Allow users to register and log in
- Allow alumni to update their profiles
- Enable administrators to manage users
- Allow posting and viewing of job opportunities
- Enable event creation and participation
- Allow students to search for alumni mentors

5.2 Non-Functional Requirements

The system shall:

- Be user-friendly
 - Be responsive on mobile and desktop devices
 - Support scalability
 - Maintain secure authentication
 - Provide fast page loading
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6. DEVELOPMENT METHODOLOGY

The project follows the Agile Software Development Methodology.

Agile Phases

Requirement Gathering

Information about alumni challenges and communication gaps was collected.

System Planning

Project structure, technologies, and architecture were defined.

Design

Wireframes, UI layouts, and database structures were created.

Development

Frontend components were developed using:

- Vite.js
- Tailwind CSS
- JavaScript

Testing

The system was tested for:

- Functionality
- Responsiveness
- Usability
- Performance

Deployment

The system is prepared for deployment on cloud hosting platforms.

7. SYSTEM ARCHITECTURE

7.1 Architecture Overview

The system follows a Client-Server Architecture.

Components

Frontend Layer

Developed using:

- Vite.js
- Tailwind CSS
- JavaScript

Responsibilities:

- User interaction
- Data presentation
- Form validation
- Responsive UI rendering

Backend Layer

Handles:

- Authentication
- API processing
- Business logic

Possible technologies:

- Node.js + Express
- Flask API
- Firebase

Database Layer

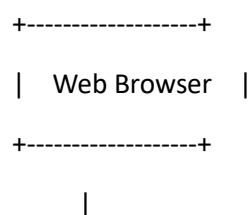
Stores:

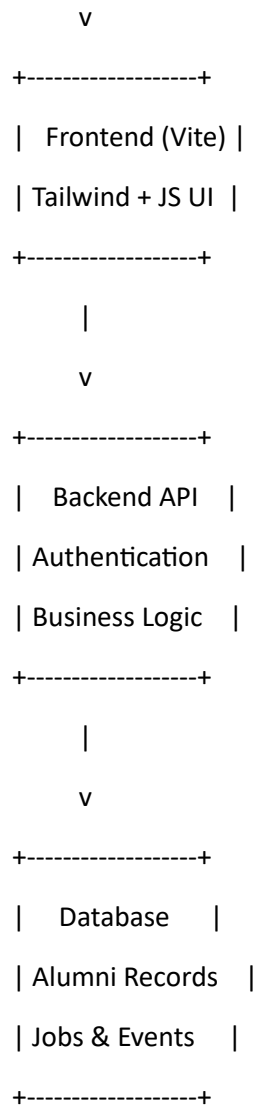
- User profiles
- Alumni records
- Events
- Job posts
- Messages

Possible databases:

- MySQL
- MongoDB
- Firebase Firestore

7.2 System Architecture Diagram





8. SYSTEM DESIGN

8.1 Use Case Design

Actors

- Alumni
- Students
- Administrator

Main Use Cases

- Register account
- Login
- Update profile
- Post jobs

- View events
- Request mentorship
- Manage users

8.2 Navigation Structure

Home

└─ About

└─ Alumni Directory

└─ Jobs

└─ Events

└─ Mentorship

└─ Login

└─ Register

9. DATABASE DESIGN

Main Tables

Users Table

Field	Type
id	Integer
fullname	Varchar
email	Varchar
password	Varchar
role	Varchar

Alumni Profiles Table

Field	Type
profile_id	Integer
user_id	Integer
graduation_year	Integer
profession	Varchar

Field	Type
company	Varchar

Jobs Table

Field	Type
job_id	Integer
title	Varchar
description	Text
posted_by	Integer

Events Table

Field	Type
event_id	Integer
title	Varchar
location	Varchar
event_date	Date

10. USER INTERFACE DESIGN

Design Principles

The interface was designed with:

- Simplicity
- Accessibility
- Responsiveness
- Modern aesthetics

Tailwind CSS Features Used

- Utility-first styling
- Responsive breakpoints
- Flexbox and Grid layouts
- Reusable UI components

Main Pages

Landing Page

Displays:

- Introduction
- Quick navigation
- Featured alumni

Dashboard

Displays:

- User profile
- Notifications
- Job opportunities
- Upcoming events

Alumni Directory

Allows searching and filtering alumni profiles.

11. FUNCTIONAL MODULES

11.1 Authentication Module

Handles:

- Registration
- Login
- Password encryption
- Session management

11.2 Alumni Management Module

Handles:

- Alumni profiles
- Career information
- Professional networking

11.3 Job Portal Module

Handles:

- Job posting
- Job browsing
- Applications

11.4 Event Management Module

Handles:

- Event announcements
- RSVPs
- Event tracking

11.5 Mentorship Module

Handles:

- Student requests
- Mentor matching
- Communication

12. DEVELOPMENT ENVIRONMENT

Software Tools

Tool	Purpose
VS Code	Code editor
Vite.js	Frontend build tool
Tailwind CSS	Styling framework
GitHub	Version control
Node.js	Runtime environment

Installation Commands

Create Vite Project

```
npm create vite@latest alumni-system
```

Install Dependencies

```
cd alumni-system
```

```
npm install
```

Install Tailwind CSS

```
npm install -D tailwindcss postcss autoprefixer
```

```
npx tailwindcss init -p
```

Run Development Server

```
npm run dev
```

13. SECURITY CONSIDERATIONS

The system implements:

- Password hashing
 - Input validation
 - Secure authentication
 - Role-based access control
 - Session protection
-

14. TESTING AND VALIDATION

Testing Methods

Unit Testing

Individual components tested separately.

Integration Testing

Modules tested together.

User Acceptance Testing

Real users tested the system usability.

Expected Results

- Fast page loading
 - Responsive layouts
 - Secure login
 - Accurate data management
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15. DEPLOYMENT STRATEGY

The system can be deployed using:

- Vercel
- Netlify
- Firebase Hosting
- Render

Deployment Steps

1. Build project

npm run build

2. Upload generated files to hosting server
 3. Configure domain and SSL certificate
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16. FUTURE IMPROVEMENTS

Future enhancements may include:

- Mobile application support
 - AI-based mentorship recommendations
 - Video conferencing integration
 - Real-time chat system
 - Alumni donation system
 - Analytics dashboard
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17. CONCLUSION

The University Alumni Management System provides an efficient and modern solution for strengthening alumni relationships and improving communication between graduates, students, and the university administration.

Using Vite.js and Tailwind CSS ensures a fast, scalable, and responsive frontend application capable of supporting future growth and additional features. The system improves alumni engagement, supports mentorship opportunities, and creates a sustainable networking platform for the university community.

The project demonstrates the practical application of modern web technologies in solving communication and management challenges within educational institutions.